YIXUAN HE

https://sherylhyx.github.io/ ⋄ Yixuan.He@asu.edu
Assistant Professor at Arizona State University

EDUCATION

University of Oxford, OU Oct. 2020 - Jun. 2024
D.Phil. (PhD) student in Statistics *Supervisors: Mihai Cucuringu and Gesine Reinert*

- **Clarendon Scholar (fully funded)**, Balliol Jason Hu Scholarship
- Dissertation title: **Graph Neural Networks for Network Analysis**

University of Edinburgh, UoE Sept.2018 - Jun. 2020
BSc (Hons) in Math and Stats *Supervisor: Timothy I. Cannings*

- GPA: **92.33%** Avg., **Rank # 1** in Math & Stats, **First Class**

University of California, Berkeley Jan. 2018 - May 2018
Semester Exchange

- GPA: **4.0/4.0**

South China University of Technology, SCUT Sept.2016 - Jun. 2020
BSc Mathematics and Applied Mathematics *Supervisor: Delu Zeng*

- Rank: **1/60** (Outstanding graduate)
- Overall GPA for 2016 - 18: **3.99/4.0**, Major GPA: **4.0/4.0**; transferred to UoE on 2+2 program
- Winning 2 years of **National Scholarship** granted by Ministry of Education of China
- **650/710** in CET6, **top 0.02 % of students** in SCUT

PROFESSIONAL POSITIONS

Assistant Professor at ASU Sept. 2024 -

- Teach statistics at School of Mathematical and Natural Sciences.
- Conduct research and mentor students.
- Organize and participate in research activities.

Applied Scientist Intern at Amazon Web Services Nov. 2021 - Aug. 2024

- Mentor: David Wipf.
- Conducted research on graph neural networks.

Game Development Engineer at NetEase Games Jun. 2020 - Aug. 2020

- Project: Behavior-based Anti-cheating in FPS Games (Supervisor: Xin Wen)
- Selected as **Outstanding Intern**.

RESEARCH EXPERIENCE

OU: Graph Neural Networks for Network Analysis Oct. 2020 - Jun. 2024
Doctoral Research *Supervisors: Gesine Reinert, Mihai Cucuringu*

- Worked on signed, directed, and temporal GNNs, as well as GNNs for signal recovery.
- Developed open-source libraries: creator of PyG Signed Directed and contributor of PyG Temporal.
- Collaborated with Amazon as an applied scientist intern (Mentor: David Wipf).

NetEase Games: Behavior-based Anti-cheating in FPS Games Jun. 2020 - Aug. 2020
Summer Internship *Supervisor: Xin Wen*

· Extracted features from logs using multiprocessing and built multi-input single-output models with TensorFlow 2.2.

UoE: AUROC vs AUPRC: A Comparison of Performance Measures in Classification Problems Sept. 2019 - Apr. 2020
UoE Honors Project *Supervisor: Timothy I. Cannings*

· Collected data, compared measure scores, **analyzed measure** and data properties.

UCLA CSST summer research/REU: Deforestation Brazil Jun. 2019 - Sept. 2019
Funded by NGA and NSF *Supervisors: Christian Parkinson, Andrea L. Bertozzi, Stanley J. Osher*

· Conducted data mining, data analysis, MATLAB programming, and model construction.
· Implemented e.g., **level set method**, numerical PDEs, optimal path searching, satellite image processing, Maximum Penalized Likelihood Estimation (MPLE), and time series simulation.
· **Delivered a talk** at IPAM's "The Level Set Collective" seminar invited by Professor Stanley Osher.
· **Presented a poster** at Intelligence Community Academic Research Symposium (ICARS) 2019.

SCUT: Robust Image Salient Object Perception Prior for Image Segmentation based on PDEs Related Methods May 2018 - May 2020
Funded by National Natural Science Foundation of China **(NSFC)** *Supervisor: Delu Zeng*

· Studied level set methods and **neural networks** on **semantic image segmentation**.
· Worked on lung nodule segmentation using MEnet.
· Utilized Linux, Colab, Caffe, TensorFlow, PyTorch, and programming in Python.

SCUT: Efficiency of Collaborative Innovation Knowledge Output in Professional Towns in Internet + Times May 2017 - May 2018
Funded by MOE of PRC *Supervisors: Shifu Wang, Miaoxi Zhao, Shenquan Liu*

· **Cross-disciplinary**: Math and Urban Planning.
· Responsible: Data mining and analysis, paper writing.
· **Led** the project as part of the National Training Program of Innovation for Undergraduates.

MAJOR AWARDS

Grant	2026 Seed Grant @ ASU Institute for Social Science Research (co-PI)
Grant	2026 Fall AI Challenge for Pilot Assist (ACPA) by Honeywell and ASU
Honor	2026 Departmental Graduate Research Excellence and Mentorship Award
Grant	2026 NVIDIA Academic Grant Program Award
Grant and Honor	2026 Jetstream2 NAIRR AI Fellows Program
Honor	2025 Departmental Exemplary Service Award
Honor	2025 Departmental Charter Award: Discovery of Public Value
Grant	2024 G-Research June Grant (mainly for attending ICML 2024)
Grant	2024 ICML Official Financial Aid
Grant	2024 ICLR Official Financial Assistance
Award	2023/2024 Academic Year Oxford University Blues Award for Sports Excellence
Grant	2024 Balliol College Support Grant
Grant	2023-24 Balliol College Floreat Grant for Sports
Grant	2022-23 Balliol College Floreat Grant for Sports
Grant	2022 ICML Official Participant Award
Grant	2022 Simons Institute Summer School Fund

Grant	2022 Balliol's fund on IMA Conf. on The Math. Challenges of Big Data registration
Award	2022 SIAM Student Travel Award for SDM'22
Award	2022 SIGIR Student Registration Award for WSDM'22
Grant	2021 Balliol's fund on Complex Networks 2021 registration
Grant	2021 Balliol's fund on NeurIPS 2021 registration
Grant	2021 Balliol's fund on ACM membership and CIKM'21 registration
Prize	2020 Napier Medal and Gangadhar Balwant Gadgil Prize (**top** Math student at UoE)
Prize	2020 Lawley Memorial Prize (**top** student in Math and Stats at UoE)
Scholarship	2020 **Clarendon Scholarship** and Balliol Jason Hu scholarship at Oxford
Honor	2020 Outstanding Graduate at SCUT Math Dept.
Prize	2019 Arthur Erdelyi Prize (For top three performing Math students at UoE)
Prize	2019 James Ward Memorial Prize (**top** student in Math and Stats at UoE)
Second Prize	2018 ICM/MCM International **Mathematical Modeling** Competition
Grand Prize	2018 Math Competition of SCUT (Math Group) (**Rank # 1**)
Honor	2017-2018 one of the 10 **Role Models to Merit Students** at SCUT
Prize	2017 and 2018 **National Scholarship**
Second Prize	2017 "9th National College Student **Math Competition** (Math)" in Guangdong
Second Prize	2017 National College Student Mathematical Modeling Competition (Guangdong)

PUBLICATIONS

* Means equal contribution. Non-arXiv publications are peer-reviewed. "Talk" means additional conference presentations.

- Sandel, A., **He, Y.**, ..., & Mitani, J. Lethal Conflict Following Group Fission in Wild Chimpanzees. (**Science**, 2026)
- Chen, Y., Wang, Z., & **He. Y.** From Link Prediction to Linear Community Detection: A Finite-Sample Guarantee for Graph Pretraining. (**NeurIPS 2026**)
- Mei, Z., Pan, R., Chen, Y., & **He, Y.** Signed Graph Pre-Training and Prompt Learning. (**Learning on Graphs Conference (LoG) 2026**)
- Zhu, J., Chen, Y., Wang, P., **He, Y.**, Li, P., Akella, A., & Wang, Z. When Do Graph Foundation Models Transfer? A Data-Centric Theory. (**ICML 2026**)
- Liang, P., Yang, C., **He, Y.**, Pan, R., & Chen, Y. Topology-Guided Graph Pre-training and Prompt Learning on Directed Graphs. (**TMLR**, 2026)
- Wang, X., Zhao, Z., Bai, K., Meng, S., Shen, D., Chang, X., & **He, Y.** TARQ: Tail-Aware Reconstruction Quantization for Rare-Word Robust Automatic Speech Recognition. (**EMNLP Findings**, 2026)
- Jing, C., Verma, A., Bera, K., **He, Y.**, & Lee, K. Amortizing Physics-Informed Neural Solvers via Graph Hypernetworks. (**LoG 2026**)
- Abolhasani, M., Ba, Y., **He, Y.**, & Pan, R. Traceable Context-Enriched Knowledge Graph Generation with Induced Schemas. (**LoG 2026**)
- Pizarro, T., Su, J., **He, Y.***, & Kang, Y. SCAR dynamics of adolescent substance use: peer influence, dropout, and bifurcation structure in a school-based model. (**CSIAM Transactions on Life Sciences**, 2026)
- **He, Y.**, Su, J., & Kang, Y. Longitudinal and Graph-Augmented Prediction of Adolescent Substance Use Onset in the ABCD Study. (AIAS+ 2026)
- Yu, H., **He, Y.** EnKiteDDI: Literature-Enhanced Drug-Drug Interaction Event Prediction. (AIAS+ 2026)
- Lu, Y., Liu, X., ..., **He, Y.**, Li, X., & Hao, Y. Position: Synthetic Persona Needs Explicit Grounding and Standardized Reporting. (COLM 2026 Workshop Social Sim)
- Liu, Y., ..., **He, Y.**, Hao, Y., & Li, X. PersonaEval: Persona-Based User Simulation for Evaluating Interactive Applications. (COLM 2026 Workshop Social Sim)

- Ng, S., ..., **He, Y.**, Li, X., & Hao, Y. MicroVerse: An Instrument for Measuring Self-Authored Identity Drift in Long-Horizon Multi-Agent Language-Model Simulations. (COLM 2026 Workshop Social Sim)
- Li, X., Hao, Y., ..., **He, Y.**, ..., & Song, D. MatrAIx: Simulating the World with 8.3 Billion Persona Agents. (arXiv 2026)
- Zhao, Z., Wang, X., Chang, X., & **He, Y.** From Mathematical to Executable Certificates for Machine Unlearning. (arXiv 2026)
- Wang, X., Zhao, Z., **He, Y.**, Chang, X., & Smola, A. When Is Deletion Ordering Tractable? From Update Dynamics to Permutation Structure. (arXiv 2026)
- Yilmaz, N., Patel, M., Kusumba, A., **He, Y.**, & Yang, Y. MentalBlackboard: Evaluating Spatial Visualization via Mathematical Transformations. (arXiv 2026)
- Zhu, L., **He, Y.**, Cucuringu, M. Quantifying the Contributions of Clustering to Statistical Arbitrage. (SSRN 2026)
- Huang, J., Fan, Y., Shen, H., ..., **He, Y.**, ..., Li, X., & Hao, Y. AgentPersonaBench: Benchmarking Persona-Driven User Simulation (arXiv 2026)
- Ockerman, S., Gueroudji, A., Mallick, T., **He, Y.**, Pouchard, L., Ross, R., & Venkataraman, S. PGT-I: Scaling Spatiotemporal GNNs with Memory-Efficient Distributed Training. (**SC25**)
- **He, Y.**, Sandel A., Wipf D., Cucuringu M., Mitani J.& Reinert G. Learning to Fuse Temporal Proximity Networks: A Case Study in Chimpanzee Social Interactions. (arXiv 2025)
- Aminian, G.*, **He, Y.***, Reinert, G., Szpruch, Ł., Cohen, SN. Generalization Error of Graph Neural Networks in the Mean-field Regime. (**ICML 2024**)
- **He, Y.**, Reinert, G., Wipf., D., & Cucuringu, M. Robust Angular Synchronization via Directed Graph Neural Networks. (**ICLR 2024**)
- Steach, H., Viswanath, S., **He, Y.**, Zhang, X., Ivanova, N., Hirn, M., Perlmutter, M., & Krishnaswamy, S. Inferring Metabolic States from Single Cell Transcriptomic Data via Geometric Deep Learning. (**RECOMB 2024**)
- **He, Y.**, Zhang, X., Huang, J., Rozemberczki, B., Cucuringu, M., & Reinert, G. PyTorch Geometric Signed Directed: A Software Package on Graph Neural Networks for Signed and Directed Graphs. (**LoG 2023**, **Open-Source Software with 100+ stars**)
- Geng, H., Chen, C., **He, Y.**, Zeng, G., Han, Z., Chai, H., Yan, J. Pyramid Graph Neural Network: A Graph Sampling and Filtering Approach for Multi-scale Disentangled Representations. (**KDD 2023**, Talk)
- Wu, Q., Yang, C., Zhao, W., **He, Y.**, Wipf, D., & Yan, J. DIFFormer: Scalable (Graph) Transformers Induced by Energy Constrained Diffusion. (**ICLR 2023**, Spotlight Talk)
- **He, Y.**, Reinert, G., & Cucuringu, M. DIGRAC: Digraph Clustering Based on Flow Imbalance. (**LoG 2022**)
- **He, Y.**, Perlmutter, M., Reinert, G., & Cucuringu, M. MSGNN: A Spectral Graph Neural Network Based on a Novel Magnetic Signed Laplacian. (**LoG 2022**)
- Wang, X., Chen, S., **He, Y.**, Wang, M., Gan, Q., & Yan, J. CEP3: Community Event Prediction with Neural Point Process on Graph. (**LoG 2022**)
- **He, Y.**, Gan, Q., Wipf, D., Reinert, G. D., Yan, J., & Cucuringu, M. GNNRank: Learning Global Rankings from Pairwise Comparisons via Directed Graph Neural Networks. (**ICML 2022**, Talk)
- **He, Y.**, Reinert, G., Wang, S., & Cucuringu, M. SSSNET: Semi-Supervised Signed Network Clustering. (**SDM 2022**, Talk)
- **He, Y.** GNNs for Node Clustering in Signed and Directed Networks. (**WSDM-DC 2022**, Talk)
- Zhang, X., **He, Y.**, Brugnone, N., Perlmutter, M., & Hirn, M. MagNet: A Neural Network for Directed Graphs. (**NeurIPS 2021**)
- Rozemberczki, B., Scherer, P., **He, Y.**, Panagopoulos, G., Riedel, A., Astefanoaei, M., ... & Sarkar, R. PyTorch Geometric Temporal: Spatiotemporal Signal Processing with Neural Machine Learning Models. (**CIKM 2021 best paper award**, **Open-Source Software with 2.9k stars, 400+ forks**)

· **He, Y.**, Hu, T., & Zeng, D. Scan-flood Fill (SCAFF): An Efficient Automatic Precise Region Filling Algorithm for Complicated Regions. (**CVPR 2019** workshop)
· Zeng, D., **He, Y.**, Liu, L., Chen, Z., Huang, J., Chen, J., & Paisley, J. Ro-SOS: Metric Expression Network (MEnet) for Robust Salient Object Segmentation. (arXiv 2018)

OTHER SELECTED ORAL PRESENTATIONS

· LIONS Seminar, Arizona State University, 2026
· Mathematical Biology Seminar, Arizona State University, 2026
· Department of Epidemiology and Biostatistics Program Seminar Series, University of Arizona, 2025
· Meta Research Seminar, Meta, 2025
· Learning on Graphs and Geometry seminar, OU, 2025
· UCSC Statistics Seminar, University of California, Santa Cruz, 2025.
· Cornell CS Theory Tea, Cornell University, 2025.
· Research Innovation in the Mathematical Sciences Seminar, ASU, 2024.
· CAM / DoMSS Seminar, ASU, 2024.
· Statistics Seminar, ASU, 2024.
· CNRS GdR IASIS Graph Learning Day, CNAM Paris, 2024.
· Statistical learning on LARge scale GRaphs (LARGR), Inria Lille - Nord Europe, 2023.
· 4th IMA Conference on The Mathematical Challenges of Big Data, OU, 2022.
· The 10th Int. Conf. on Complex Networks and their Applications, Spain (virtual), 2021.

TEACHING EXPERIENCE

ASU Assistant Professor Sept. 2024 -

· Grad course: Statistics for Bio Data Sci II (F24)
· Undergrad courses: Regression and Time Series Analyses (F25, F26), Stat Analysis for Researchers (SP25, SP26), Nonparametric Statistics (SP25, F26), Statistical Computing (SP26)

OU Teaching Assistant Dec. 2020 - Dec. 2022

· Courses to grade and discuss a problem in the problem classes: Applied Statistics (F22), Advanced Topics in Statistical Machine Learning (SP22), Computational Statistics (SP21)
· Course to demonstrate and answer questions: Statistics and Data management (for DTC in Dept. of Zoology, F20)

UoE Digital Skills Trainer - Python Dec. 2019 - May 2020

· **Delivered Python training courses**, reviewed and improved existing Python training.
· Recorded lectures: Introduction to Python and Python for Data Science.

Teaching Support in a Primary School in Longsheng Rural Village July 2017

· Devised and delivered lectures in singing, dancing, craftsmanship, and storytelling.
· **Led** in hosting the closing ceremony.
· **Visited families** to collect information and offer suggestions.

ACADEMIC SERVICES

Area Chair (AC)/Reviewer

- Conferences: WWW 2027, AAAI 2027, ACL ARR 2026 May, NeurIPS 2026, GFM at ICML 2026 (AC), ICML 2026, ICLR 2026, AISTATS 2026, AAAI 2026, NeurIPS 2025, ICML 2025, AISTATS 2025, ICLR 2025, AAAI 2025, NeurIPS 2024, ICML 2024, ICLR 2024, AAAI 2024, NeurIPS 2023, ICML 2023, WWW 2023, AAAI 2023, NeurIPS 2022, SIGKDD 2022, ICML 2022, WSDM 2022
- Journals: IEEE Transactions on Knowledge and Data Engineering, IEEE Transactions on Neural Networks and Learning Systems, Nature Scientific Reports, IEEE Transactions on Signal and Information Processing over Networks, Journal of Complex Networks, Journal of Quality Technology, Electronic Journal of Statistics, Statistics and Computing

(Co-)Organizer

- Learning on Graphs Conference 2026, international conference hosted in Boston (program chair)
- ASU Machine Learning Day 2026, ASU
- Learning on Graphs Conference 2025, international conference hosted at ASU (local host)
- ASU Machine Learning Day 2025, ASU
- AI Seminar at ASU, ASU
- Learning on Graphs Local Meetup at Arizona 2024, ASU
- LoG^2 at Oxford: the Learning on Graphs and Geometry seminar series, OU

CONFERENCES/SUMMER SCHOOLS

40th Conf. on Neural Information Processing Systems **(NeurIPS)**	Author	Dec. 2026
The Inter. Conf. for HPC, Networking, Storage, & Analysis (SC26)	Expo Presenter	Nov. 2026
AIAS+	Presenter & Author	Nov. 2026
Agentic AI and the Student Experience 2026	Panelist	Oct. 2026
43rd International Conference on Machine Learning **(ICML)**	Author & GFM Panelist	Jul. 2026
2026 AAAS Annual Meeting – Science at Scale	Volunteer	Feb. 2026
4th Learning on Graphs (LoG) Conference	Organizer and Local Host	Dec. 2025
41st International Conference on Machine Learning **(ICML)**	Author	Jul. 2024
CNRS GdR IASIS Graph Learning Day	Speaker	Jun. 2024
12th International Conference on Learning Representations (**ICLR**)	Author	May 2024
2nd Learning on Graphs (LoG) Conference	Author	Nov. 2023
Workshop: statistical learning on LARge scale GRaphs (LARGR)	Presenter	Mar. 2023
1st Learning on Graphs (LoG) Conference	Author	Dec. 2022
4th IMA Conference on The Mathematical Challenges of Big Data	Speaker	Sept. 2022
39th International Conference on Machine Learning **(ICML)**	Author	Jul. 2022
SIAM International Conference on Data Mining **(SDM)**	Author	Apr. 2022
15th ACM Int. Conf. on Web Search and Data Mining **(WSDM)**	Doctoral Consortium	Feb. 2022
35th Conf. on Neural Information Processing Systems **(NeurIPS)**	Author	Dec. 2021
The 10th Int. Conf. on Complex Networks and their Applications	Author	Nov. 2021
Int. Conf. on Information and Knowledge Management **(CIKM)**	Author	Nov. 2021
Intelligence Community Academic Research Symposium **(ICARS)**	Poster Presenter	Sept. 2019
Computer Vision and Pattern Recognition **(CVPR)**	Author	Jun. 2019

COMPUTER SKILLS

Programming:	**Python (Advanced)**, C++ (Intermediate), VB (Basic), and Pascal (Basic)
Statistical Analysis:	**R (Advanced)**, SQL (Intermediate), and SPSS (Intermediate)

Modelling:	**Matlab (Advanced)**, Maple (Intermediate), and Xpress (Intermediate)
Paper writing:	**Latex (Advanced)**

POSITIONS OF RESPONSIBILITY

President of Oxford University Table Tennis Club (OUTTC) May 2023 - May 2024

- **Organized** club activities, promoted the club, organized training, managed booking, and coordinated facilities.
- Participated in table tennis **competitions** as a squad member.
- **Initialized** the use of an Oxford mailing list to send notifications (600+ subscribers) and **managed** a club Facebook group (1.2k members).

Director of Dept. of Visiting Scholars in Oxford Chinese Students and Scholars Association (OXCSSA) July 2021 - July 2022

- **Co-organized** 2021 welcoming event for new students to OU.
- **Organized** events for visiting scholars: e.g., visiting scholars' tea party, hiking, formal, and punting.
- Collaborated with other departments to hold virtual and in-person talks.

Head of Class Sept. 2013 - Jun. 2018

- Conveyed notifications, organized activities, and led by example.
- 2017 and 2018 South China University of Technology **Outstanding Student Leader**.
- 2016-2017 **outstanding class** of SCUT and 2017-2018 top-16 classes of SCUT.

University Student Representative May 2017 - May 2018

- Actively listened to students$'$ opinions and participated in raising proposals.
- Outstanding Student Representative of South China University of Technology in 2017.

Member of Public Relationship Dept., Math Student Union Sept. 2016 - Oct. 2017

- **Led** the hosting of a social party in School of Mathematics.
- 2017 **Star** in Mathematics Student Union.